

DEVELOPER BOARD TRENDS

For Industrial Applications



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Hrishikesh Kamat, CEO, Shalaka Technology, recalls the challenges faced by one of his clients due to their machine operators' lack of understanding of a machine. The CNC machine, which cost over ₹ 100,000, broke down and plant administrators and operators could not figure out the fault in time. Unfortunately, the company had to replace the costly CNC machine.

Had there been an industrial board instead of the regular board in the CNC machine, it would have indicated to the administration specific breakdown points, ensuring long-term efficient functioning of the machine with no room for errors.

Industrial boards are regular design boards that embed multiple functions in one single unit or board, allowing users to interface smoothly with any application as per their needs.

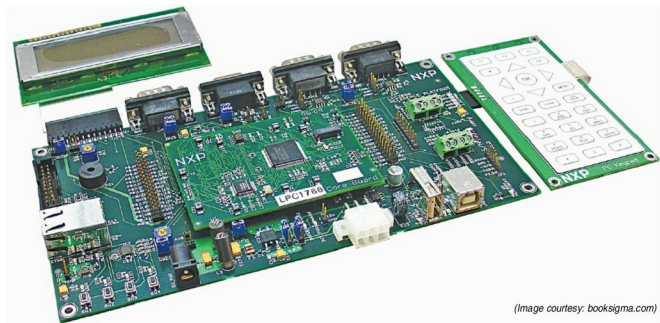
Understanding the need for industrial boards

From robotics to embedded systems, factory automation to building automation,

smart grid to motor drives, smart lighting to space, avionics and defense, industrial developer boards are significant in almost every aspect. These are quite handy for designers as they need not rework the PCB layout for different applications, with just the controller programming serving the purpose. Industrial developer boards are customer-centric as against regular developer boards, which are more product-centric in design.

For instance, a regular power supply has inbuilt over-voltage, over-current and over-load protection. But if an industrial application demands an alarming signal along with blinking of an LED to indicate over-voltage, over-current or over-load conditions, there will be a specified industrial developer board to serve this purpose. This board will then be modified by either using a microcontroller circuit to program the IC or by making changes in the components based on the calculations done.

In microcontroller-based industrial boards, functions can be added by



(Image courtesy: booksigma.com)